Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cylinders and spheres, and volume. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	5
Topic	Three dimensional shapes II
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume
Theme	—
Curriculum objectives	• identify properties of cylinders and spheres • find the volume of a cube, a cuboid and a prism
Previous knowledge	• properties of cubes and cuboid • properties of pyramids and cones
Lesson sequence	• identify properties of cylinders and spheres — the teacher displays shapes of cylinders and spheres; the student identify the properties; the teacher draws the shapes on the board and adds to the properties already mentioned by the students. Set for the students: 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces. • find the volume of a cube, a cuboid and a prism — the teacher explains how to derive the formula for volume of cube and cuboid, and works the volume of a prism as area of base times height; the student listen and ask questions where they do not understand; the student copy worked examples into their notes
Instructional materials	• shapes of cylinders and spheres.
Evaluation	• Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high. • Calculate the volume of cuboid with dimensions 5m by 2m by 3m. • What is the volume of a cube of 4cm × 4cm × 4cm. • A rectangular tank hold 6m long by 2m wide holds $36m^3$ of water. How deep is the water in the tank.

Lesson objectives

1.	identify properties of cylinders and spheres
2.	find the volume of a cube, a cuboid and a prism

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cylinders and spheres, and volume

Class: JSS1 **Subject:** Mathematics **Week:** 5

Section 1: Cylinders and Spheres: Knowing Their Shapes

You have already worked with shapes like cubes, cuboids, pyramids and cones. Now we meet two more solid (three-dimensional) shapes: the cylinder and the sphere. A solid shape is described by its faces, edges and vertices, so to identify a shape we count and name these parts carefully.

Start with the **cylinder**. Think of a tin of baked beans, a drinking straw, or a roll of toilet paper. A cylinder has:

- **Two circular faces** – one at the top and one at the bottom. These two circles are exactly the same size, and they lie flat and parallel to each other.
- **One curved surface** – the smooth, rounded wall that joins the two circles together.
- In total, a cylinder has **three faces**: two circular faces plus the one curved surface.
- A cylinder has **no vertices** – you will not find a sharp corner or point on it.
- A cylinder has **no straight edges** either, because the circular rims are curved, not straight.

So when someone asks how many faces a solid cylinder has, the answer is **three**: two circular faces and one curved surface.

Now think of a **sphere**. Picture a football, a marble, or an orange. A sphere is perfectly round – every point on its surface is the same distance from its centre. A sphere has:

- **Exactly one curved surface**, and no flat faces at all.
- **No edges** and **no vertices** – there is nothing sharp anywhere on it.

So the key difference between the two shapes is this: a cylinder has flat circular faces (two of them) and a curved surface, while a sphere is one smooth curved surface with no flat faces, no edges and no vertices.

Worked example 1 – a cylinder

Look at a solid cylinder. Let's count its parts step by step.

1. Hold the cylinder so you can see the top. There is a circular face. Flip it over – there is another circular face at the bottom. That gives **2 circular faces**.
2. Run your hand around the wall that joins the top to the bottom. That smooth rounded wall is the **curved surface** – **1 curved surface**.
3. Add them together: 2 circular faces + 1 curved surface = **3 faces** in total.
4. Finally, look for corners. A cylinder has **no vertices**, and it has **no straight edges**.

So a solid cylinder has **3 faces** (2 of them circular) and **no vertices**.

Worked example 2 – a sphere

Take a sphere, like a football.

1. Feel the whole surface. It is smooth and curved all over, with no flat part – so it has **1 curved surface** and **no flat faces**.
2. Look for any corner, edge or point. There are none – a sphere has **no vertices** and **no edges**.

So a sphere has just **1 face** (a curved one) and **no vertices**.

The two worked examples show the difference clearly: the cylinder uses two flat circular faces plus a curved surface, while the sphere is one smooth curved surface with nothing flat and nothing sharp.

Practice

1. How many circular faces does a solid cylinder have?
2. A cylinder has how many faces in total?
3. How many faces, edges and vertices does a sphere have?
4. Explain one key difference between a cylinder and a sphere.

Section 2: Finding the Volume of Cubes, Cuboids and Prisms

Volume tells you how much space a three-dimensional shape takes up. To measure it, think of filling the shape with little cubes. For a cuboid, you can picture it as rows of unit cubes.

A cuboid has three measurements: its length, its width (also called its breadth) and its height. To find its volume, you multiply all three together:

Volume of a cuboid = length × width × height

A cube is a special cuboid where the length, width and height are all equal. If the side of the cube is called s, then its volume is:

Volume of a cube $= s \times s \times s = s^3$

So a cube of side 4 cm has volume $4 \times 4 \times 4 = 64$ cm^3.

A prism is a three-dimensional shape with two identical parallel faces (its bases) and sides that are rectangles or parallelograms. For any prism, you can find its volume by working out the area of the base first, then multiplying by the height of the prism:

Volume of a prism = area of base × height

A cuboid is itself a prism whose base is a rectangle, so this same rule works for it too: the area of its rectangular base is length × width, and multiplying by the height gives length × width × height.

Remember the units: if the measurements are in centimetres, the volume is in cubic centimetres, written cm³; if in metres, it is cubic metres, m³.

Example 1: Volume of a cuboid

Find the volume of a cuboid that is 8 cm long, 4 cm wide and 3 cm high.

Volume = length × width × height
= 8 × 4 × 3
= 32 × 3
= 96 cm³

Example 2: Volume of a cube

What is the volume of a cube of side 5 cm?

Volume $= s^3 = 5 \times 5 \times 5$
= 25 × 5
= 125 cm³

Example 3: Volume of a prism

A triangular prism has a base with area 10 cm² and a height of 7 cm. Find its volume.

Volume = area of base × height
= 10 × 7
= 70 cm³

Practice

1. Calculate the volume of a cuboid that is 8 cm long, 4 cm wide and 3 cm high.
2. Calculate the volume of a cuboid with dimensions 5 m by 2 m by 3 m.
3. What is the volume of a cube of 4 cm × 4 cm × 4 cm?
4. A rectangular tank is 6 m long and 2 m wide, and it holds 36 m³ of water. How deep is the water in the tank?

Check your work

Attempt every question before you read this page.

Cylinders and Spheres: Knowing Their Shapes

1. 2 (two). A solid cylinder has two circular faces – one at the top and one at the bottom – plus one curved surface.
2. 3 faces: two circular faces and one curved surface.

   Count the flat parts first: the top circle is one face and the bottom circle is another, giving 2 faces. Then add the curved wall that joins them, which is 1 more face. So 2 + 1 = 3 faces in total.
3. A sphere has 1 face (a single curved surface), no edges and no vertices.
4. A cylinder has two flat circular faces plus one curved surface, while a sphere has only one curved surface and no flat faces, no edges and no vertices. (Either difference – flat circular faces versus none, or no edges and no vertices on a sphere – is a correct answer.)

Finding the Volume of Cubes, Cuboids and Prisms

1. 96 cm³

   Volume = length × width × height = 8 × 4 × 3 = 32 × 3 = 96 cm³.
2. 30 m³

   Volume = 5 × 2 × 3 = 10 × 3 = 30 m³.
3. 64 cm³

   Volume = $4 \times 4 \times 4 = 16 \times 4 = 64$ cm³.
4. 3 m deep

   Volume of the cuboid tank = length × width × depth, so 36 = 6 × 2 × depth = 12 × depth. Therefore depth = 36 ÷ 12 = 3 m.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-da2b8d16	**Date**	
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____